Cálculo Diferencial e Integral II Semestre: 2024/2

Nome: Thiago Dellano Machado de Oliveira – 24101272

Questão extra A2

Questão Extra

As equações de Maxwell relacionam o campo elétrico $\vec{\mathbf{E}}$ e o campo magnético $\vec{\mathbf{H}}$, quando eles variam com o tempo em uma região que não contenha carga nem corrente, como segue:

$$\boldsymbol{\nabla} \cdot \vec{\mathbf{E}} = 0$$

$$\boldsymbol{\nabla} \cdot \vec{\mathbf{H}} = 0$$

$$\boldsymbol{\nabla} \times \vec{\mathbf{E}} = -\frac{1}{c}\frac{\partial \vec{\mathbf{H}}}{\partial t}$$

$$\boldsymbol{\nabla} \times \vec{\mathbf{H}} = \frac{1}{c}\frac{\partial \vec{\mathbf{E}}}{\partial t}$$

onde c é a velocidade da luz. Use essas equações para demonstrar o seguinte:

a)

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \vec{\mathbf{E}}) = -\frac{1}{c^2}\,\frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2}$$

b)

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \vec{\mathbf{H}}) = -\frac{1}{c^2}\,\frac{\partial^2 \vec{\mathbf{H}}}{\partial t^2}$$

c)

$$\nabla^2 \vec{\mathbf{E}} = \frac{1}{c^2}\,\frac{\partial^2 \vec{\mathbf{E}}}{\partial t^2}$$

d)

$$\nabla^2 \vec{\mathbf{H}} = \frac{1}{c^2}\,\frac{\partial^2 \vec{\mathbf{H}}}{\partial t^2}$$

Dicas:

- *tome* $\vec{\mathbf{H}} = \langle h_1,\ h_2,\ h_3 \rangle$ *e* $\vec{\mathbf{E}} = \langle E_1,\ E_2,\ E_3 \rangle$*;*
- *use o teorema de Fubini;*
- *use a identidade* $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \vec{\mathbf{F}}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \vec{\mathbf{F}}) - \nabla^2 \vec{\mathbf{F}}$

2 pts adicionais na prova até o limite 10,00

a) $\nabla \times (\nabla \times F) = \nabla(\nabla \cdot F) - \nabla^2 F$

$\nabla \times (\nabla \times E) = \nabla(\nabla \cdot E) - \nabla^2 E$

$\nabla \times (\nabla \times E) = -\nabla^2 E$

$\nabla \times (\nabla \times E) = -\nabla \times \left(\frac{1}{c}\frac{\partial H}{\partial t}\right)$

$\nabla \times \left(\frac{1}{c}\frac{\partial H}{\partial t}\right) = \frac{1}{c}\frac{\partial}{\partial t}(\nabla \times H) = \frac{1}{c}\frac{\partial}{\partial t}\left(\frac{1}{c}\frac{\partial E}{\partial t}\right)$

$\nabla \times (\nabla \times E) = -\frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$

b) $\nabla \times (\nabla \times H) = \nabla(\nabla \cdot H) - \nabla^2 H$

$\nabla \times (\nabla \times H) = -\nabla^2 H$

$\nabla \times (\nabla \times H) = -\nabla \times \left(\frac{1}{c}\frac{\partial E}{\partial t}\right)$

$\nabla \times \left(\frac{1}{c}\frac{\partial E}{\partial t}\right) = \frac{1}{c}\frac{\partial}{\partial t}(\nabla \times E) = \frac{1}{c}\frac{\partial}{\partial t}\left(-\frac{1}{c}\frac{\partial H}{\partial t}\right.$

$\nabla \times (\nabla \times H) = -\frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$

c) $\nabla \times (\nabla \times E) = -\nabla^2 E$

$\nabla \times (\nabla \times E) = -\frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$

$-\nabla^2 E = -\frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$

$\nabla^2 E = \frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$

d) $\nabla \times (\nabla \times H) = -\nabla^2 H$

$$\nabla \times (\nabla \times H) = -\frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$

$$-\nabla^2 H = -\frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$

$$\nabla^2 H = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$$